

The Game Away from Home: Evaluating the Impact of Performance in Longer Distance Away Games

Kevin Lynch Larry Nguyen Alejandro Zamudio

Abstract

With the schedule and travel demands of collegiate level sports becoming much heavier for the average athlete, concern grows over the impact it may have on their individual performance. In this project, our group attempts to answer if travel distance impacts the performance of the Texas Longhorns Softball team and their players. Our group approaches this question by looking at both average team performance and individual player performance. By looking at both, we are able to evaluate if a correlation with distance exists between how players perform and how the team as a whole performs. We also implement the use of a win probability model using a logistic regression, aimed at evaluating which circumstances of travel was the team most likely to win. Our findings ultimately highlight to us that distance alone may not be a great predictor of team performance and that players are impacted on an individual basis. We then summarize our findings and put our findings into the context of the sport and continuously changing demands of college sports.

Research Question and Objective

In a dynamic environment for college sports, many schools find their athletic teams traveling longer distances to play against higher levels of competition. A great example of this is reflected in College Softball. Every season, many top teams travel far distances to play in invitationals, conference games, and regional tournaments in the pursuit of reaching the Softball College World Series and winning a National Championship. With the potential of having to

travel multiple times in a single week, players can potentially find themselves exhausted and overwhelmed from the demands of their schedule along with the sport and school. While numerous research has evaluated the overall effects that long distance travel has on athletes (specifically physical and mental health), our investigation takes a more centered approach on performance within the sport rather than the physical and off the field sphere. With this in mind, our research question centers around the impact travel distance has on performance in College Softball, and if longer travel distance can be attributed to poorer performances.

Our investigation will center around the Texas Longhorns Softball team and their recent play in the 2022 season to the 2024 season. With the focus centered on in-game performance, our main objective is to identify if the team performed better or worse in longer distance games. To better guide our investigation, our group developed two focus questions. First, did the team tend to perform better when designated as the home team or the away team? It is important to note that by “performance”, we are interested in more than just record and runs scored per game. We want to look at metrics on both the hitting and pitching side of the game to note any differences between home and away games. Furthermore, we use the term “designate” as we factor in Neutral Site games where the Longhorns may have been the home or away team (games in which home field advantage would not apply to either team). Second, is there any noticeable trend in performance in away games that are played in further locations from the team’s home venue? By factoring in distance of the game from the home venue when looking at specific hitting and pitching metrics, we can attempt to evaluate if it has any notable correlation with performance. By also factoring in other confounding variables that could impact performance, we can also determine if distance works as a variable alone or if other variables play into the impact distance has.

In this paper, we start off by providing the context of the data used by our group and our methodology for how we intend to approach the investigation. Next, our group then provides an analysis of the team's initial performance as both the home and away team. By getting a baseline understanding of how the Longhorns performed in the three seasons, we can use the context to better explain any findings and results later in the investigation. We then move into identifying the distance to away venues and evaluate notable differences and correlations in performance. In that section, we explain how we calculated distance and the use of other variables when evaluating performance. We also introduce the use of a Win Probability model aimed at understanding the likelihood of the team winning particular games in certain circumstances. Lastly, we summarize our findings and give context in the new era of College Sports. We also note areas for improvement in our investigation.

Data Collection and Preparation

For our investigation, we utilized three different sources of data. The first source of data was box score and schedule datasets from SoftballR, an R Package. The packages provided us Hitting and Pitching box scores for all of college softball for the timeline we set, along with a schedule for the games played and results in these three seasons (*see appendix for dataset column descriptions*). The hitting datasets contained variables such as the name of the players, the amounts of hits, walks, strikeouts and other batting metrics in a standard box score. For pitching, we are given the individual pitcher, the amount of innings they pitched in the game, and additional metrics such as strikeouts, walks and runs allowed. Following filtering the dataset to only include games played by the Longhorns, we did have to go in and manually fill in game and box score data to all three datasets as a total of eight games were missing. Furthermore, we

added additional data metrics to the data such as batting average, slugging, and ERA for our performance-based metrics. In terms of the schedule, we added additional columns to represent game type (tournament or neutral site), the runs Texas scored, and the distance from their home venue (we will expand on this in the methodology).

The second source of data was a database of RPI rankings for each individual team in Division 1 Softball by D1Softball. The purpose of the dataset is to help us put a ranking to the associated opposing team the Longhorns played on their schedule. The RPI rankings considers wins against teams of various rankings and strength of schedule when formulating its final ranking of the specific team. The reason we wanted to include this into our analysis is because the strength of the teams played by the Longhorns provide huge context to how they might've played in certain circumstances or at certain distances. Playing tougher competition can result in games with possibly poorer performance in comparison to a game against a weaker opponent. By including the associated RPI ranking of each team, we can understand the quality of opponents being played and if that correlates as well with distance. The distance variable is included into the schedule dataset and is also intended to be used in our Win Probability model (will be expanded on in methodology).

Lastly, our final source were post game box game reports that were available on the team's website. Stored in PDF format, we downloaded each pdf for every game in order to gather information about the location of the game and as well as fill in any missing information in the data provided by the SoftballR package. Overall, the postgame summaries simply served to help validate the data we have and provide us the venue details we aimed to use for our analysis and machine learning model.

As mentioned, the RPI rankings of each team were added as an additional column to the schedule dataset for our R package. The hitting, pitching and schedule dataset serve as the main dataset locations for our analysis. Furthermore, we also performed an inner join of the schedule dataset with the pitching and hitting datasets so that we separate datasets that better showed the performance associated with the specific game without accidentally altering the original datasets. All results of the investigation were provided by analysis done on the three datasets mentioned.

Exploratory Data Analysis

In our initial exploratory analysis of the data, our group wanted to get an idea of the overall performance by the Longhorns over the three season span to help give us an understanding of the team's baseline performance. Before looking at home and away games individually, we wanted to get an idea of the performance of the team as a whole. In the three season span, the Longhorns played a total of 196 games with a record of 145 wins and 51 losses. 105 games took place in invitationals, Regionals and Super Regionals, and College World Series games. However it is also important to note that a majority of tournament games took place at the teams home venue (we count these as home games for future reference). Averaging roughly 6 runs per game and allowing roughly 3 runs in this time frame, the team consistently won in high margins of runs scored and dominated their opponents defensively (2024 season saw the highest in the three year span with roughly 7 runs scored per game). The average RPI ranking of their opponents in all three respective seasons was 52, 57 and 55 with them having played the top ranked team in all three seasons.

Next, we split the games based on whether the team was marked as the home or away team. We additionally split the home and away games into “True” games to also factor in and

evaluate non neutral site games. When designated as the home team, the Longhorns had a record of 92-26. As the designated away team, the team went 53-25. While the team did play fewer games as the away team, the trend of playing better at home was well reflected in each individual season in which the team had a far higher win percentage at home with the exception of 2024 (*See Table 1*). Furthermore we can also see that in “True Home and Away” games, the team was still better at home in terms of win percentage. However, this could be likely due to the fact that the team played higher ranked opponents on average on the road compared to at home (40, 50 and 45 average RPI of the opponent when away).

Table 1:

Season	Designated Home	Designated Away	True Home	True Away
2022	27-13	19-11	23-10	17-8
2023	33-8	11-9	30-6	7-7
2024	32-5	23-5	26-4	18-5

Looking at offensive metrics, we performed a similar analysis to get a good idea of how well the team performed. When looking score wise, our group found that the team scored slightly more runs designated as the home team than the away team with the exception of 2024 (*See Figure 1*). Furthermore, when splitting the offensive metrics of the team as a whole based on home and away games, we can further back up that team were more efficient at hitting the ball and putting up runs in the games as batting average and runs seems to reflect a huge difference between the two (*See Table 2*). It is important to note that the numbers are also influenced by the larger amount of games played while designated as the home team. However with the strikeout

and walk numbers being closer in certain seasons, we do see some consistency in that hitting at home was a strength for this Texas team.

Figure 1:

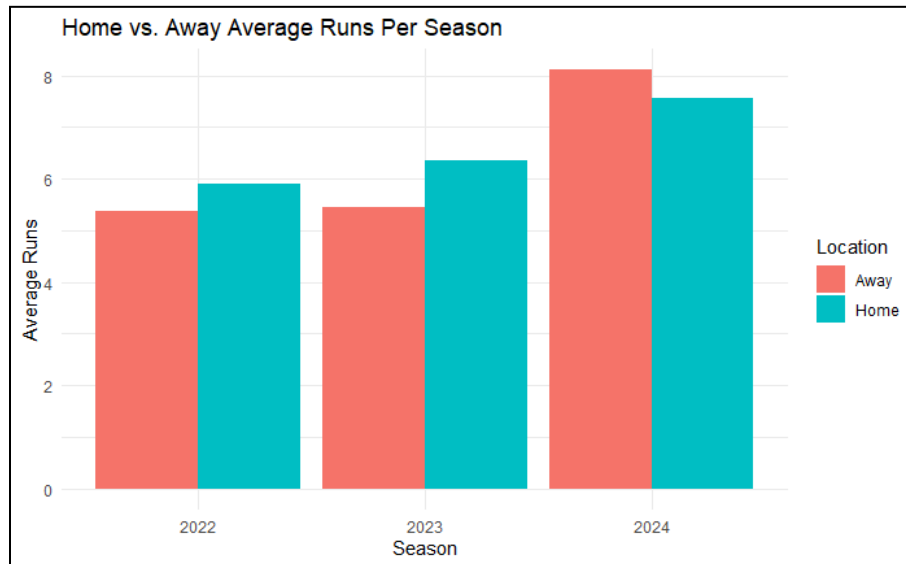


Table 2:

Designation	BA	SLG	K	BB
2022 - Home	0.319	0.509	150	138
2022 - Away	0.284	0.468	147	85
2023 - Home	0.333	0.506	141	110
2023 - Away	0.308	0.448	79	61
2024 - Home	0.375	0.646	129	135
2024 - Away	0.363	0.569	94	112

Looking at the pitching side of games, we do get a similar trend in that the pitching staff found itself performing better when designated as the home team with the exception of a single

year. **Figure 2** shows that with the exception of the 2022 season, the team sported a fairly lower ERA when designated the home team. Similar to the hitting, we also went further by looking at the Pitching specific stats based on the seasons. What we noticed is that in some metrics, the team performed better on the road such as runs allowed and hits given up. However the team still had higher strikeouts at home and fairly similar walks in both (*See Table 3*). Again, we figure that some of these numbers are large in difference due to the different amounts of games played designated as each.

Figure 2:

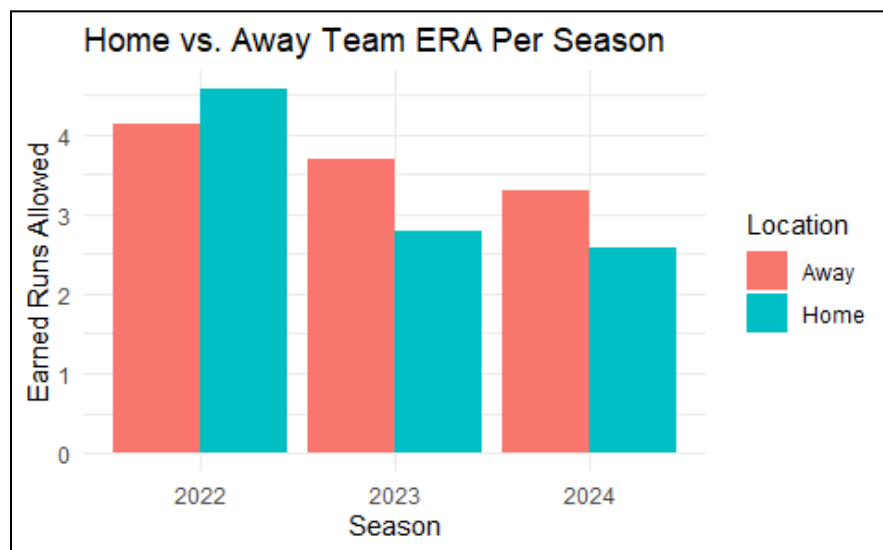


Table 3:

Designation	HA	ER	K	BB
2022 - Home	237	129	255	102
2022 - Away	187	86	195	62
2023 - Home	223	81	230	75
2023 - Away	114	51	88	45
2024 - Home	198	70	183	62
2024 - Away	131	63	121	57

Lastly, we evaluated the teams performance in Neutral Site and Tournament specific games. In Neutral Site, the team sported records of 6-6, 6-4 and 9-1, highlighting early struggles in games where they faced high competition (39, 56 and 46 average RPI in these games over the three seasons). In terms of pitching and hitting, the first two seasons saw some struggle with ERA over 3.5 and batting average no more than .300. However, 2024 was the best season in terms of neutral site games.

Overall, our exploratory analysis seems to reflect that in all three seasons, the team seemed to perform way better than the home team (whether neutral site or not). The team consistently found itself hitting better and pitching more dominantly in home games in comparison to the away. The performance at home also came with a stark difference with the away games. While the team put up fairly good numbers as the away team, there still seems to be a huge difference with how they played in comparison to at home. With this analysis aimed at providing context for our possible results, the tendency to have tougher performances when the

away team might provide the possibility that distance could be a factor in how well the team played.

Methodology and Results

During our investigation, our group will look at two specific areas of performance: performance on a team average basis and performance on an individual player basis. With these two areas of analysis, we are able to observe how the team as a whole performed in further distance games while also noting any particular trend for specific players. We can also put into context how each individual player's performance may affect the team averages and reflect what the numbers say about the team. We go even further by also splitting our scope of games into "Individual" games and "Series" sets of games. The analysis of Individual games only looks at one game as its own data point while our series sets look at average performance by the team or player throughout the specific tournament or conference series played. The reason we included the series set is to eliminate the effect poorer single game performance could have on our analysis. Players have the occasional bad day, so we want to see how they looked throughout the entire weekend or tournament as well. In the end, we had a total of 72 series marked in our data which included tournaments, one game non conference games, regionals and college world series games.

To begin, we first calculated the distance (in miles) of every game in relation to how far they were from McCombs Field (The team's home venue). Using the website "mapdevelopers.com", an online tool that computes distances given two addresses, we calculated the driving distance using the address of McCombs Field, and set the end address as the away venue. The reason for choosing driving distance is that it provides us a consistent measurement

of distance and possible time it might take to travel to an opposing venue as opposed to possible flight and ground travel times. Using the postgame report files for each game, we identified the away stadium and performed all calculations in our schedule dataset in R. All games that were played at McCombs Field, including tournament and regional matches, were marked with a 0 while the values for away venues were rounded to 2 decimal places. In our calculations of distance, we found that the team traveled an average distance of 864 miles in 2022, 721 miles in 2023 and 656 miles in 2024.

Starting off with average team performance, we wanted to first evaluate the correlation of many hitting and pitching metrics with distance. By doing this, we could identify metrics of interest and see if we can highlight any notable performance improvements or drops as distance from home increases. One of the biggest observations we made during this step was that for both the individual game and series data, a majority of the metrics available saw very little correlation with distance with many of the correlation coefficients falling under .10. This included hitting metrics such as team walks, hits and batting average, as well as pitching metrics such as strikeouts and earned runs (ER). However, when factoring out home games and only including games and series that took place in away venues, we began to notice slightly higher, however still unconvincing correlations. Two examples can be seen via the amount of times the team struckout (hitting), and the average ERA of the pitchers playing in the game / series (pitching). On an individual game basis, strikeouts and distance carry a correlation of 0.17 while the series basis has a correlation of 0.42 with a much more noticeable visual trend in our graph of the metric (**Figure 3**). For hits allowed on the pitching side, an opposite effect can be observed where on a series basis, hits allowed had a correlation of 0.07 while the individual game basis saw a -0.15 correlation with distance (**Figure 4**).

Figure 3:

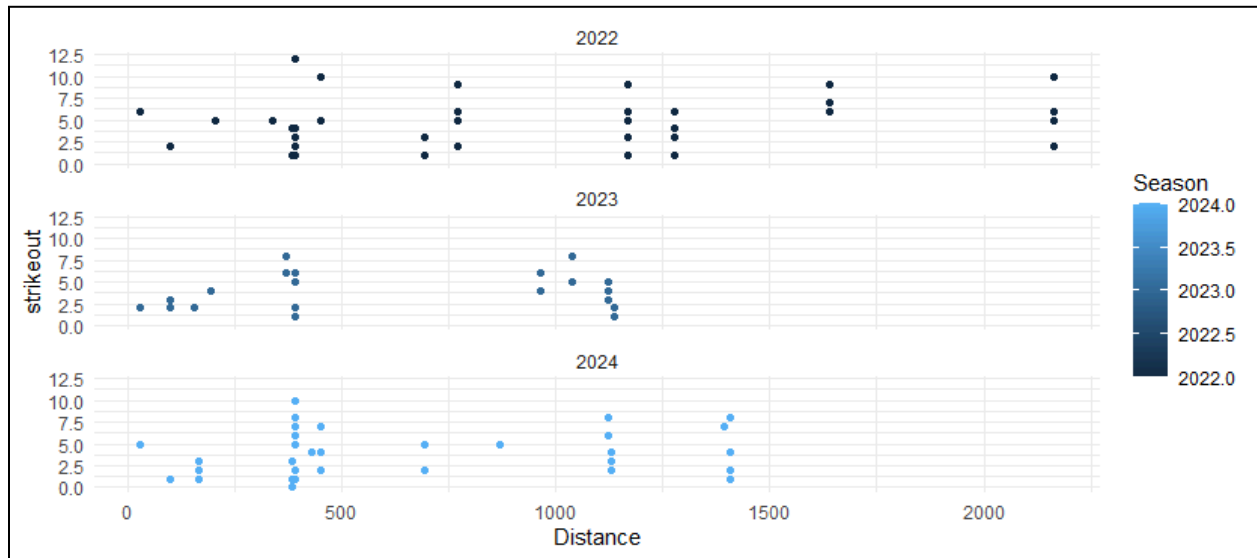
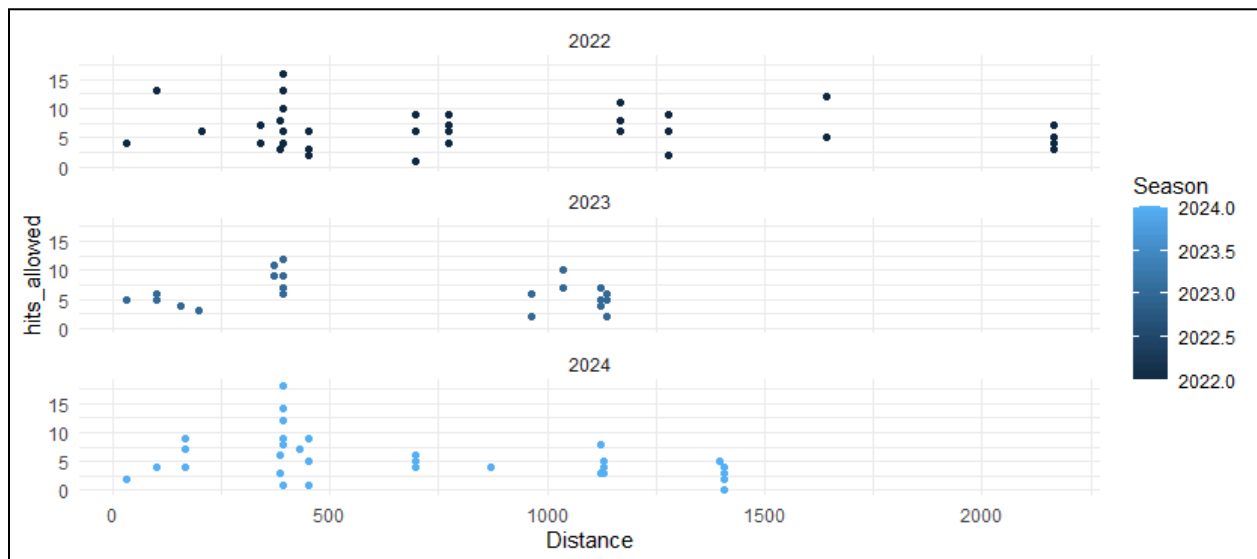


Figure 4:



When just observing the metrics without considering home games, we found that both sets of data for individual games and series had higher correlations with distance in comparison when we include data from the home games. However, very few of these correlations went over .20 and had very little visual indication that distance could be a factor. While the sample size of how many points we have does play into this, especially when grouping games by series, we still find very few relationships between the performance metrics and distance.

Next, we began analyzing the correlation of metrics with distance for each individual hitter and pitcher on the team. For this specific analysis, we only looked at performance on a series since individual game performance can look heavily different depending on how well the player has recently played. Furthermore, we separated game performance based on a range of distance: Home games at 0, closer games at less than 500 miles, and further games at greater than 500 miles. The reason for this approach was that we felt it could better reflect individual player performance, especially with the fact that not every player plays the same amount of games or gets the same amount of at bats / innings. We would have a better sample size across each player to make better judgments about their performance at certain distances of away locations. What we found is that all players, whether a pitcher or hitter, seemed to have fairly different performances with no consistency in better or worse performance at any distance. An example on the hitting side can be seen in 2024 where outfielder Isabella Dayton sported a higher batting average in closer games (0.371) in comparison to further away games (0.290). This is in high difference with teammate Reese Atwood who seemed to excel much more in further games (0.430 BA in comparison to 0.316). We also noted that returning players often at times had differing performance season to season as one hitter with a high on base in further distance games could have a much lower on base the next year in the same classification of

distance. For the pitching side, one example was pitcher Estelle Czech who had a way lower ERA in further away games (2.25 in 2022, 1.25 in 2023 and sub 0 in 2024) in comparison to much closer games (Over 4 ERA in all 3 seasons). The pitching did not reflect similar differences year to year for returning players, however no pitcher had similar success to each other at similar classifications of distance.

Our results seem to reflect differences on a player to player basis. No two hitters or pitchers tend to perform the same at certain distances and often have some fluctuation in their performance the following year if they are returning. Many played better in games closer to home and some significantly better when played further away from home. As mentioned earlier in our exploratory analysis, the team tended to play much tougher competition on the road when it came to RPI rankings. While the players obviously had better metrics at home, the fluctuation in performances at certain distances could be a result of the various tougher competition played. The effects could possibly be seen in regional and world series games, possibly affecting the correlation at certain distances where the games are being played.

Lastly, we developed a win probability model with the intent to see if a logistic regression could provide us insight into which distance games the team is likely to win. Using the classification of intervals of distance used in the players analysis, we fed our model attributes such as the amounts or runs the team scored, the opponent RPI rank, distance of the game from the home venue, days since the previous game and trained it on whether the team won that game (classifier). Our model then returned a decimal that represents the likelihood the team won that specific game according to the model. Putting the resulting values in our schedule dataset, we see that the team was more likely to win in further distance games (Greater than 500 miles) than in shorter distance games in the 2023 and 2024 seasons. However, the 2022 season shows the

opposite effect with a higher win probability in closer games. Home games reflected the highest out of all the classifications (besides the 2024 season where it was the same as the further distance probability). One thing to note is that the 2022 and 2023 season saw much lower ranked opponents on average compared to 2024, making it very likely the teams probability would be higher against the opponents played (**Table 4**).

Table 4:

Season	Home	Short Distance	Long Distance
2022	0.757	0.654	0.623
2023	0.774	0.622	0.773
2024	0.827	0.67	0.827

Note: Short Distance: Away games within 500 miles. Long Distance: Away games further than 500 miles

Conclusion

As travel continues to become a consistent norm for those in Collegiate Softball, distance alone may not be an independent factor that impacts how these teams perform. From the results of our investigation, our group saw that distance alone and in the context of the longhorn's performance rarely correlated with any form of improvement or struggle in performance. In our analysis of the team's average performance in certain hitting and pitching metrics, we found very little correlation with distance to indicate whether the team often played better or worse at certain distances. Furthermore, our following analysis of individual players then highlighted that success and struggle at certain distances tended to differ from player to player. While we would assume most players, especially when travelling closer distances, would consistently perform better due to less travel stress, our results showed that rarely was the case. Even with the longhorns

constantly facing tougher opponents in further games, certain players seemed to just excel way beyond their performance in closer distance games.

In the context of college athletics as a whole, our results hint at the idea that long distance travel may not be as exhaustive for current collegiate athletes; and that other external factors for an athlete would need to be evaluated to see if their performance is affected by travel distance. Especially at a time where high level athletes do lots of physical and mental preparation, they are well conditioned for the demands that come with travelling and competing for their sport. Indicators of recent physical health decline such as injuries or aspects of physical and mental preparation for competition that involves heavy travel could better determine if long distance travel impacts an athletes or teams performance (rather than metrics and stats from their sport). As our results showed, performance at various travel distance varies player to player; so it is likely preparation and situations that affect performance away from home would differ from player to player as well

One limitation we felt may have impacted the results collected by our group is that we only looked at one team over a relatively short period of time. While we did consider three consecutive seasons, looking at one team may not be enough data to find meaningful relationships. By including other teams and considering a longer time frame of games, we could understand how numerous teams are impacted by distance traveled and see if we can find similar or differing results to the ones found for this specific team. Furthermore, other statistics such as fielding percentages and more physical aspects like player speed running the bases, hitting exit velocities and pitch speeds might give us better insight into more in depth aspects of performance. These types of details could better allow insight into how consistently the team is performing on a physical level and if travel has any impact on it. Lastly, our win probability

model might have been limited in the predictors we used and even the type of machine learning model. Adding more predictors like the teams batting average and ERA in those specific games or other details surrounding the conditions could improve the insights.

Appendix

Dataset 1: Schedule (2022-2024)

Note: Italicized indicated columns added during our project

- Away_team (string): name of the away team
- Away_team_runs (int): runs scored by the away team
- Home_team (string): name of the home team
- Home_team_runs (int): runs scored by the home team
- Game_date (datetime): date the game was played
- Season (int): year in which the game was played
- Game_id (int): number identifier for the particular game
- *Series (int): number identifier for games in a series*
- *Days_since_prev (int): days since the previous game*
- *Neutral_site (binary): binary representation for if the game is neutral site*
- *Tournaments (binary): binary representation for if the game is part of a tournament*
- *Distance_miles (int): distance in miles from the home game*
- *Opponent_rpi_rank (int): opposing teams associated RPI rank*
- *Texas_win (binary): binary representation for if Texas won the particular game*

Dataset 2: Hitting (2022-2024)

Note: Italicized indicated columns added during our project

- Player (string): name of the players associated with the displayed box score
- Pos (string): abbreviation of the players position for the particular game
- G (int): Games played
- Rbi (int): Runs batted In
- Ab (int): At Bats
- R (int): Runs
- H (int): Hits
- X2b (int): Doubles
- X3b (int): Triples
- Tb (int): Total bases
- Hr (int): Home Runs
- Ibb (int): Intentional Walks
- Bb (int): Walks
- Hbp (int): Hit by Pitch
- Sf (int): Sacrifice Fly
- Sh (int): Sacrifice Hit

- **K (int): Strikeout**
- **Kl (int): Strikeout Looking**
- **Dp (int): Double Play**
- **Gdp (int): Grounded into Double Play**
- **Tp (int): Triple Play**
- **Sb (int): Stolen Base**
- **Cs (int): Caught Stealing**
- **Picked (int): Picked off on the bases**
- **Go (int): Ground Out**
- **Fo (int): Flyout**
- **Team (string): name of the team**
- **Opponent (string): name of the opponent**
- **Game_id (string): number identifier for the particular game**
- **Game_date (datetime): day the game was played**
- **Season (int): year the game was played**
- *Ba (float): Batting Average*
- *Slg (float): Slugging Percentage*
- *Ops (float): On base plus slugging*

Dataset 3: Pitching (2022-2024)

Note: Italicized indicated columns added during our project

- **Game_id (string): number identifier for the particular game**
- **Team (string): name of the team**
- **Opponent (string): name of the opponent**
- **Player (string): name of the player**
- **Ip (int): Innings Pitched**
- **Ha (int): Hits Allowed**
- **Er (int): Earned Runs**
- **Bb (int): Walks**
- **Hb (int): Hit Batters**
- **So (int): Strikeouts**
- **Bf (int): Batters faced**
- **Hr_a (int): Home Runs Allowed**
- **Go (int): Ground Outs**
- **Fo (int): Flyouts**
- **Season (int): year the game was played**

